

Prospective source model of the 2026 M_w 6.8 Kumamoto (Japan) earthquake

A blind test of the MTUQ+WISP pipeline, produced before the USGS finite fault

Automated pipeline (17.Venezuela_2026)

Frozen 2026-07-28 13:19 UTC, hours after origin — USGS posted their finite fault 3.2 h later (16:33 UTC); the head-to-head comparison in Section 6 was added after the freeze

Purpose

No USGS finite-fault model existed for `us6000tgb9` when this was produced. We run the teleseismic pipeline on open GSN/FDSN data and freeze the prediction so it can be compared with the USGS (and JMA/NIED) solutions once published. Only teleseismic body + surface waves and the public W-phase moment tensor (as seed) are used — no strong motion, no GNSS, no InSAR.

1 Event and frozen benchmark

Origin	2026-07-28T07:27:15.5 UTC, 32.682°N, 130.722°E, depth 10 km		
Magnitude	M_w 6.8 (W-phase); $M_0 = 1.73 \times 10^{19}$ N m; centroid 11.5 km; DC 87.6 %		
Tectonics	shallow crustal strike-slip, Kumamoto fault zone (2016 M_w 7.0 sequence region)		
W-phase NP1	212.5/75.1/−163.8	right-lateral, NE–SW Futagawa–Hinagu trend	
W-phase NP2	118.2/74.3/−15.5	left-lateral conjugate	

2 Data

The rewritten fetcher delivered the best coverage of any event in this project: 241 three-component stations (641 channels), 32–89°, maximum azimuthal gap 30°. WISP applied its own quality control on top; MTUQ used response-corrected subsets of 25 and 60 stations.

M6.8 Kumamoto teleseismic stations (n=240; rings 30/60/90 deg)

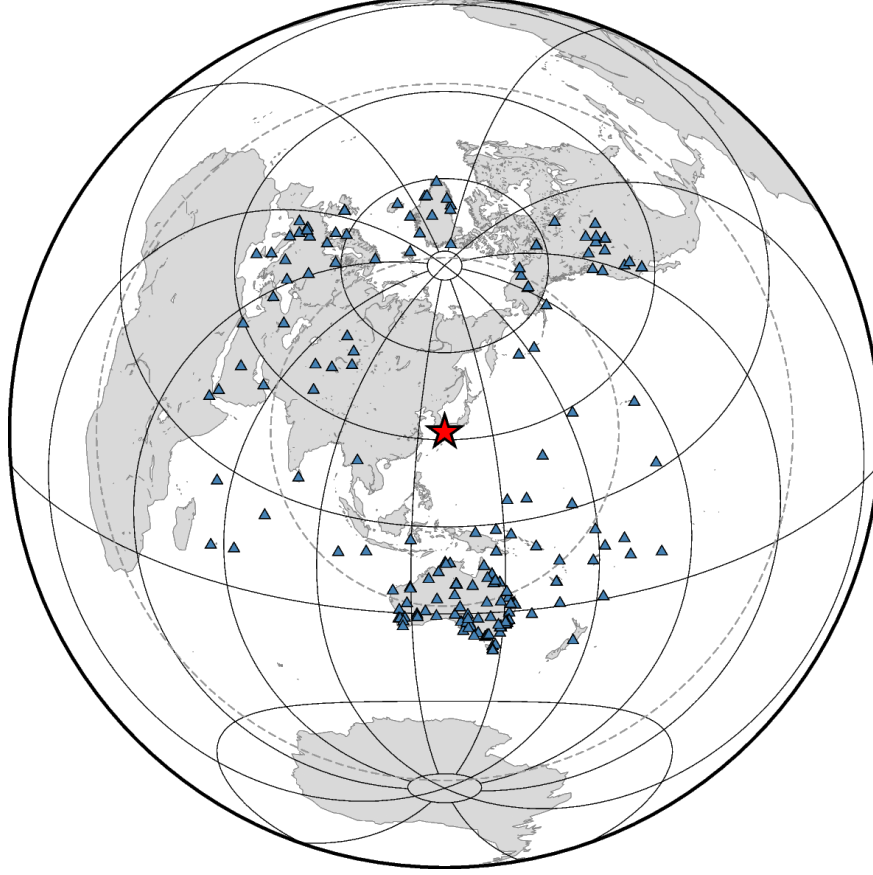


Figure 1: Teleseismic stations (azimuthal-equidistant; rings 30/60/90°).

3 Finite fault (WISP), both nodal planes, **shift_match** applied

WISP `auto_model` on both W-phase planes, seeded from the W-phase moment, followed by the cross-correlation time-shift step and re-inversion (standing rule; it improved the misfit by 9 % and 6 %).

plane	strike/dip/rake	M_w	misfit (pre→post shift)	peak slip	T_{5-95} / V_r
NP2 = Futagawa trend	212/75/−164	6.75	0.1607 → 0.1507	1.88 m	13.6 s / 2.05
NP1 = conjugate	118/74/−16	6.75	0.1667 → 0.1523	1.63 m	14.8 s / 2.09

Table 1: The tectonically expected right-lateral plane fits marginally better ($\sim 1\%$). As in our Morocco/Chile/Mexico studies the teleseismic fit is a weak plane discriminant; here the weak preference at least points the same way as the regional tectonics.

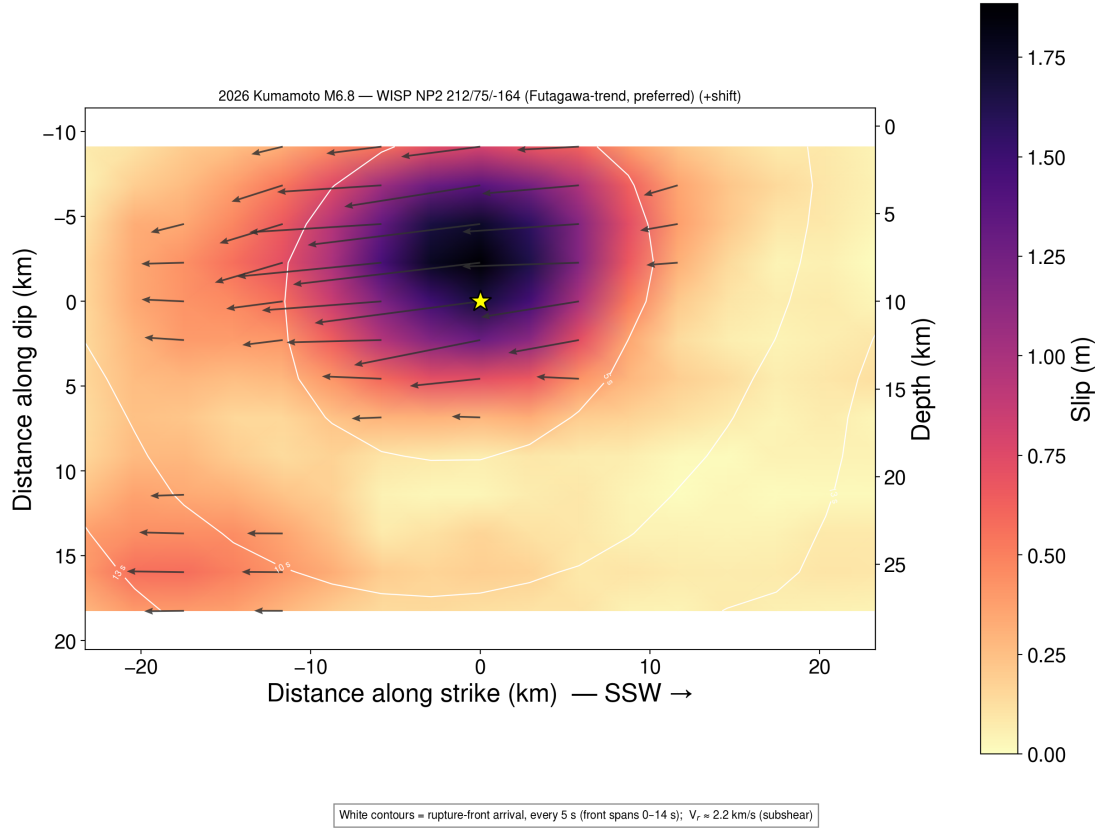


Figure 2: Preferred plane (212/75/-164): single compact asperity slightly updip of the 10-km hypocentre, slip ~ 2 –16 km depth, peak 1.88 m, rupture front spanning ~ 14 s at $V_r \approx 2.1 \text{ km s}^{-1}$ — a smaller sibling of the 2016 Kumamoto rupture.

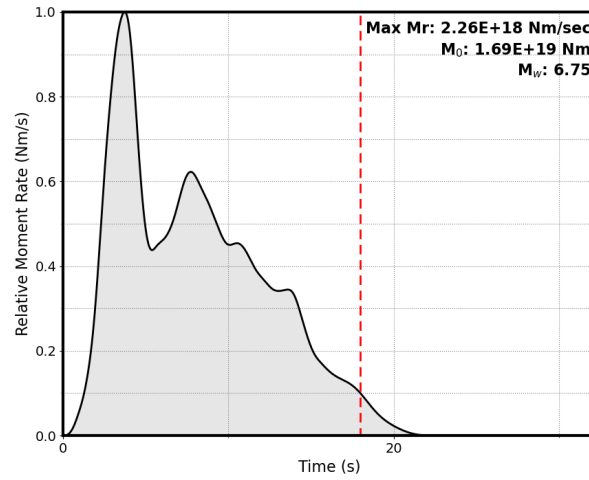


Figure 3: Moment-rate function (preferred plane).

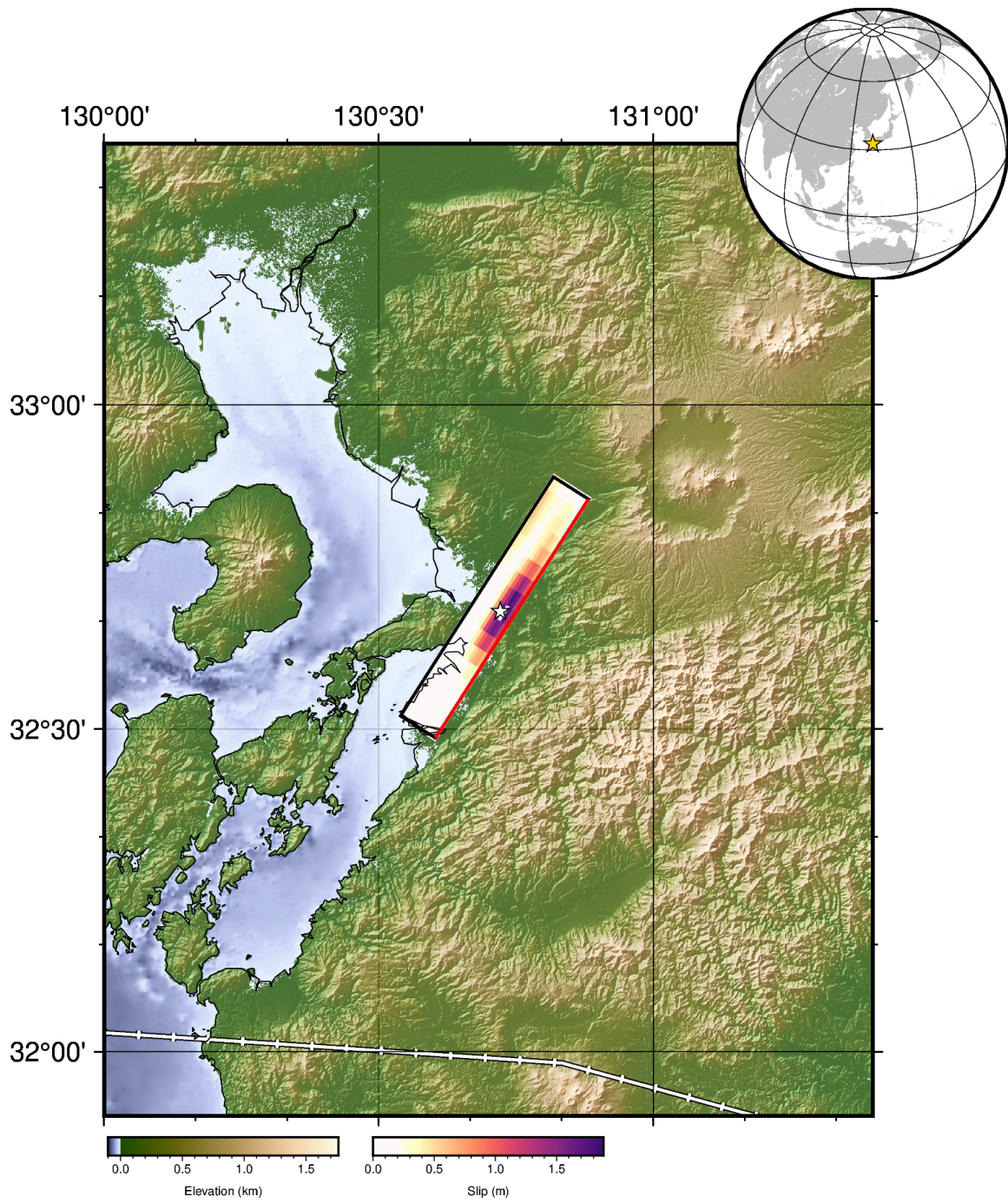


Figure 4: Map view of the preferred-plane slip.

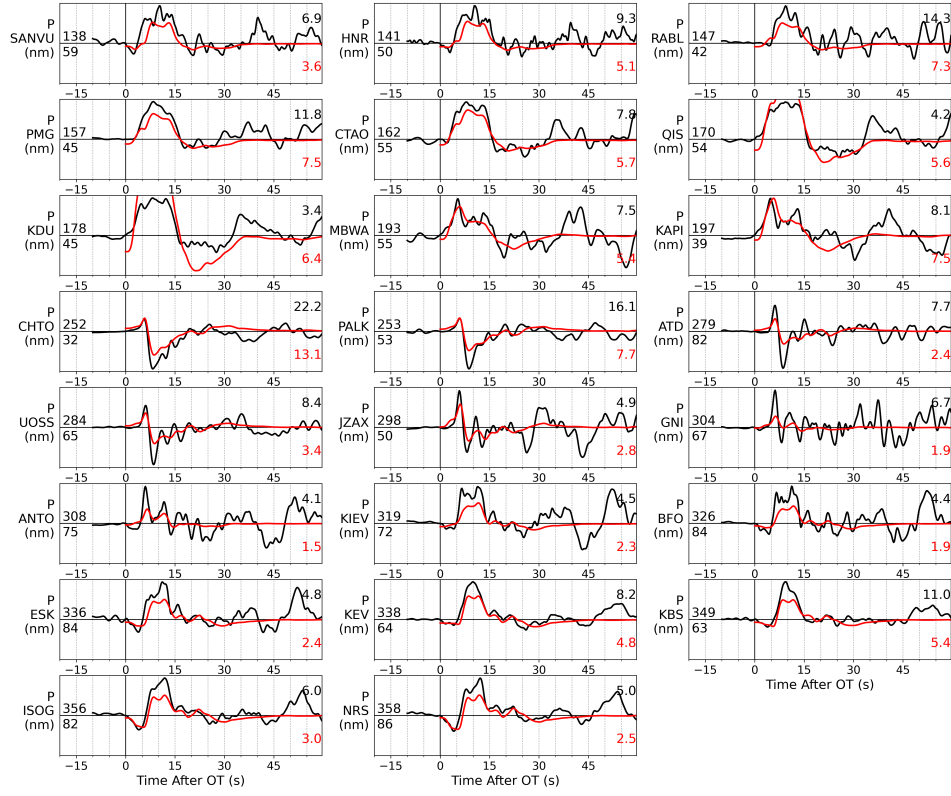


Figure 5: Teleseismic P fits (preferred plane, shifted).

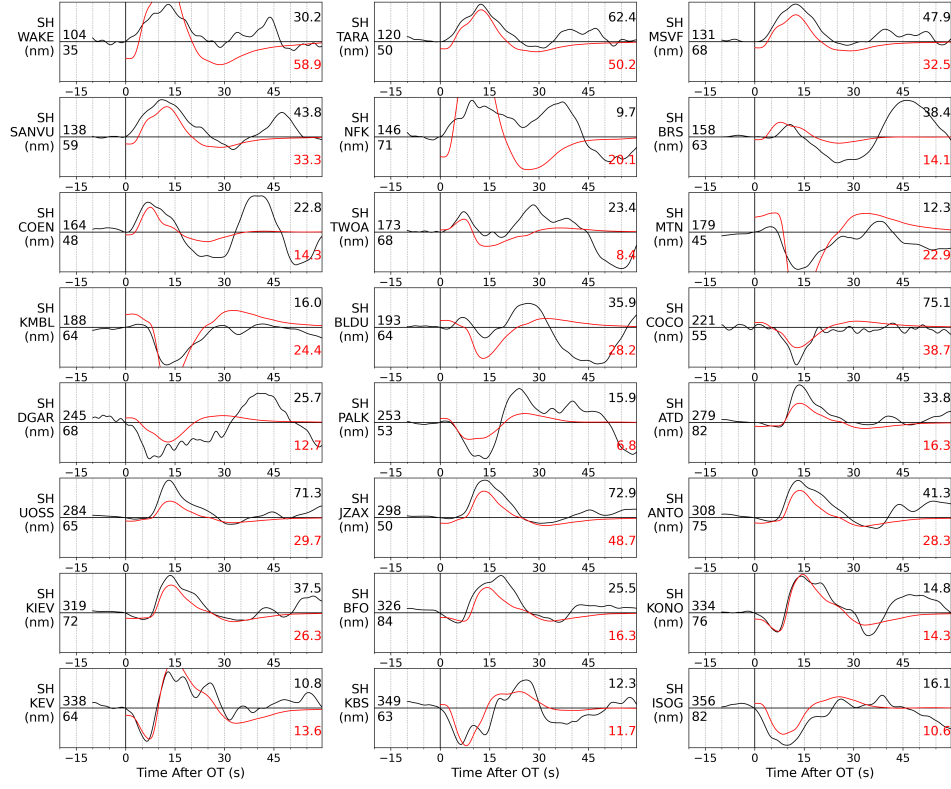


Figure 6: Teleseismic SH fits (preferred plane, shifted).

4 Moment tensor (MTUQ) — an honest failure mode, documented

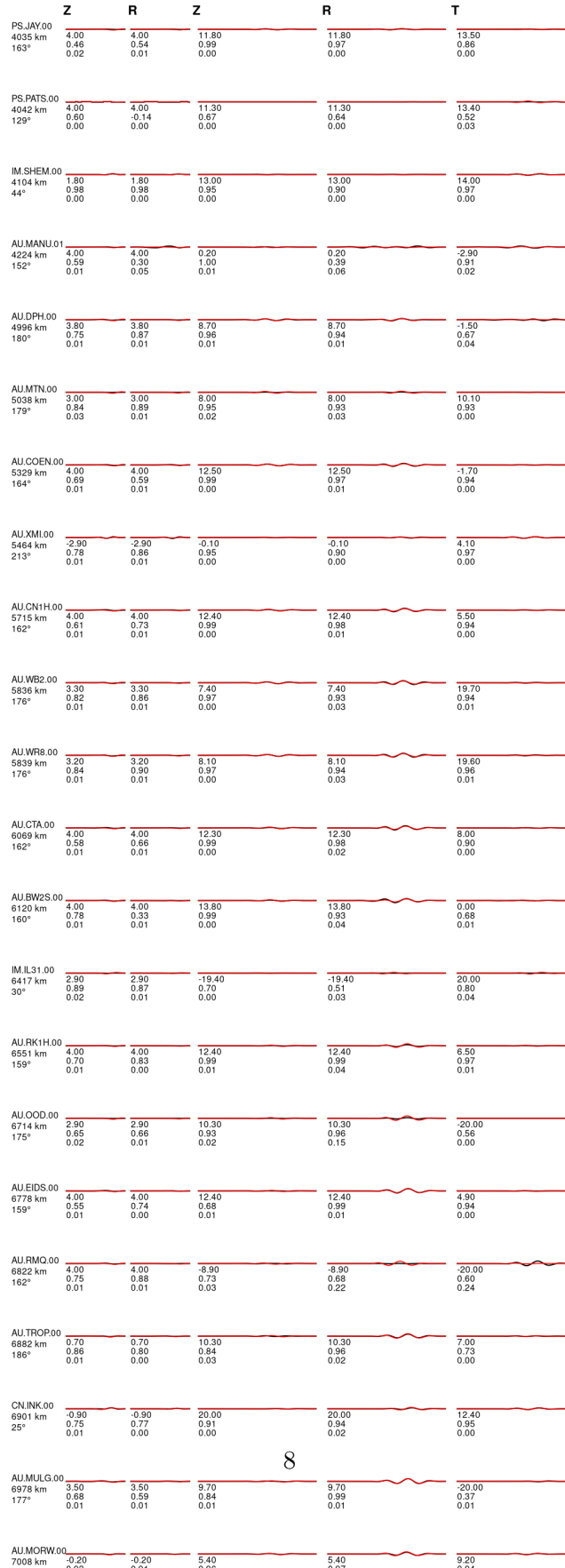
The independent MTUQ mechanism estimates were unstable for this event, in an instructive way:

variant	stations / gap	mechanism	M_w	Kagan vs W-phase
DC, body P only	25 / 97°	94/59/−29	(edge)	27.1°
DC, body P+SH	25 / 97°	58/50/−83	6.30	67.1°
DC, joint	25 / 97°	184/82/11	6.65	~78°
full MT (surface)	25 / 97°	198/81/9 ($\gamma=\delta=0$)	6.60	78.0°
DC, joint, 60 stations	60 / 80°	130/88/7	6.80	27.3°
DC, body P+SH, 60 stations	60 / 80°	212/85/−2	6.75	91.1°

Table 2: MTUQ variants. With the 25-station subset (14 of 25 stations inside one 45° southern azimuth band) the surface-wave solutions lock onto a mechanism rotated $\sim 90^\circ$ from the W-phase — the four-lobe aliasing expected for vertical strike-slip radiation, whose patterns repeat every 90° of azimuth. Doubling the stations (gap 97° $\rightarrow$ 80°) snapped the joint solution out of the rotated minimum (78° $\rightarrow$ 27.3°) and recovered M_w 6.80 — confirming the aliasing diagnosis. The body P+SH variant remains aliased; teleseismic SH is unusable at this azimuth geometry. The pipeline’s independent mechanism precision here ($\sim 27^\circ$) is honestly below the 5–19° achieved for the thrust events.



2026-07-28T07:27:15 32.68°N 130.72°E M_w 6.80 Depth 11.5 km
model: ak135f_2s solver: syngine misfit (L2): 1.899e+00
body waves: 10.0 - 33.3 s (40.0 s), surface waves: 50.0 - 100.0 s (300.0 s)
strike dip slip: 130 88 7, lune coords γ δ : 0 0, N-Np-Ns: 60-60-60



Two things follow. First, for strike-slip events azimuthal balance matters more than for dip-slip events, because clustered sampling aliases the four-lobed radiation pattern; station count alone is not protection. Second, the shallow-event “body-only” rule from our Morocco/Mexico work is a dip-slip cure (Kanamori–Given indeterminacy of M_{rt}, M_{rp}) and does not transfer to vertical strike-slip, where the long-period surface waves are, in principle, well behaved — given adequate azimuthal sampling.

5 Frozen prospective prediction

- Fault plane: right-lateral 212/75/−164 (Futagawa–Hinagu trend), weakly preferred by the waveform fit and strongly by tectonic context.
- M_w : 6.75 (WISP, both planes), vs W-phase 6.8.
- Slip: single compact asperity, peak ~ 1.9 m, ~ 2 –16 km depth, centred at / slightly updip of the hypocentre; no shallow slip maximum at the surface.
- Duration: $T_{5-95} \approx 14$ s, moment-rate peak at ~ 3.5 s; $V_r \approx 2.1$ km s $^{-1}$.

Caveats. Teleseismic-only; the plane preference is $\sim 1\%$ in misfit (weak); the independent point-source mechanism could not be stabilised from the azimuthally clustered long-period subset (documented above), so the mechanism statement leans on the W-phase planes plus the relative WISP fit. USGS will likely add regional/InSAR data; per our Al Haouz study their model may be more compact than ours.

6 Head-to-head with the USGS finite fault (added after the freeze)

USGS (D. Goldberg) posted their model at 16:33 UTC on 2026-07-28 — 3.2 h after our frozen commit (ba7d8aa, 13:19 UTC), so the test is genuinely blind.

quantity	ours (frozen)	USGS
fault plane	212/75/−164	212/75/−164 — identical
M_w (M_0)	6.75 (1.69×10^{19} N m)	6.754 (1.70×10^{19} N m)
peak slip	1.88 m	2.01 m
slip depth extent	~ 2 –16 km	0.7–17 km ($>15\%$ of peak)
asperity	single compact, at/updip of hypocentre	single compact, ± 5 km along strike, 3–12 km
T_{5-95}	13.6 s	11.9 s
moment-rate peak	~ 3.7 s	3.4 s
V_r	2.05 km s $^{-1}$	2.46 km s $^{-1}$ (avg)
fault dimensions	50 $\times$ 30 km	60 $\times$ 31.2 km

Table 3: Every headline prediction verified: plane choice, moment, a single compact asperity at the same place and depth, duration, no shallow slip maximum. The main difference is the low-slip tail: our model puts somewhat more residual slip NE of the epicentre, USGS more SW.

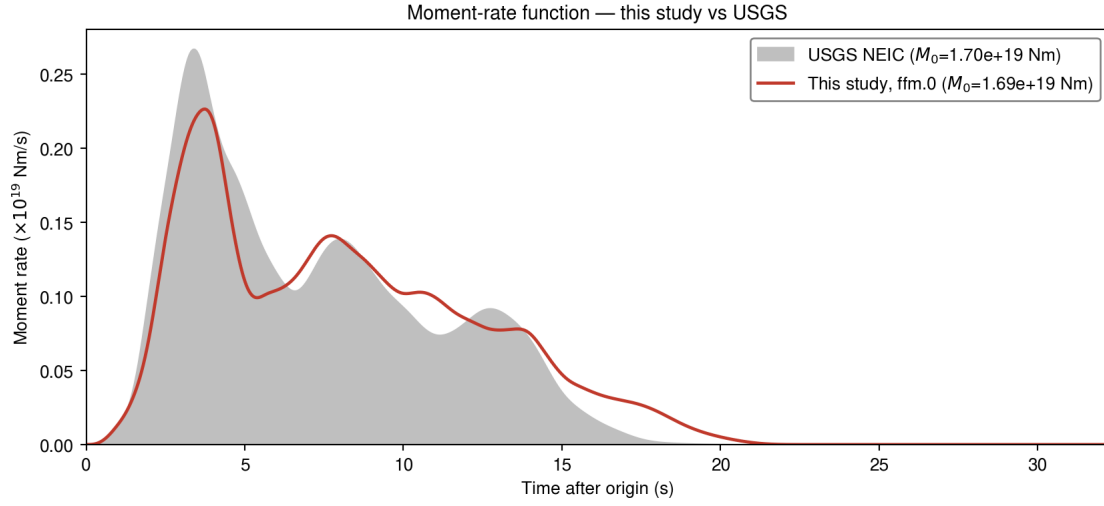


Figure 8: Moment-rate functions overlay almost perfectly — including the secondary pulses near 8 s and 13 s.

Map-view slip distribution — this study vs USGS

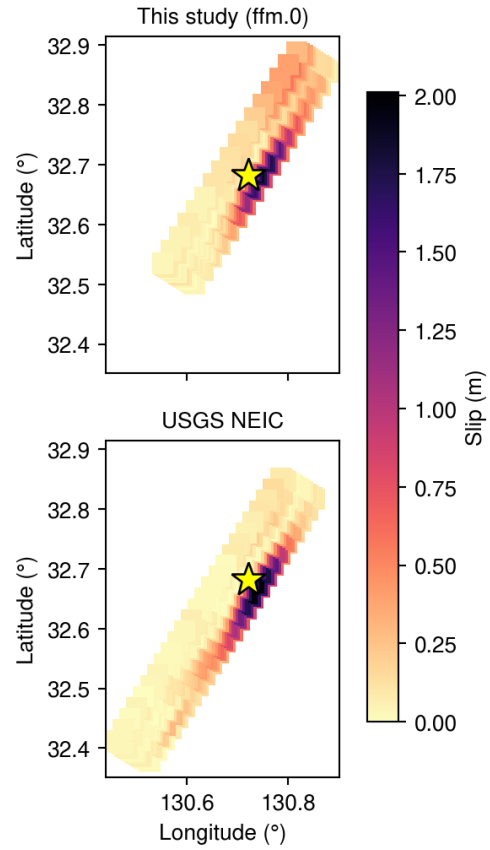


Figure 9: Map-view slip: the same NE–SW high-slip streak in both models.

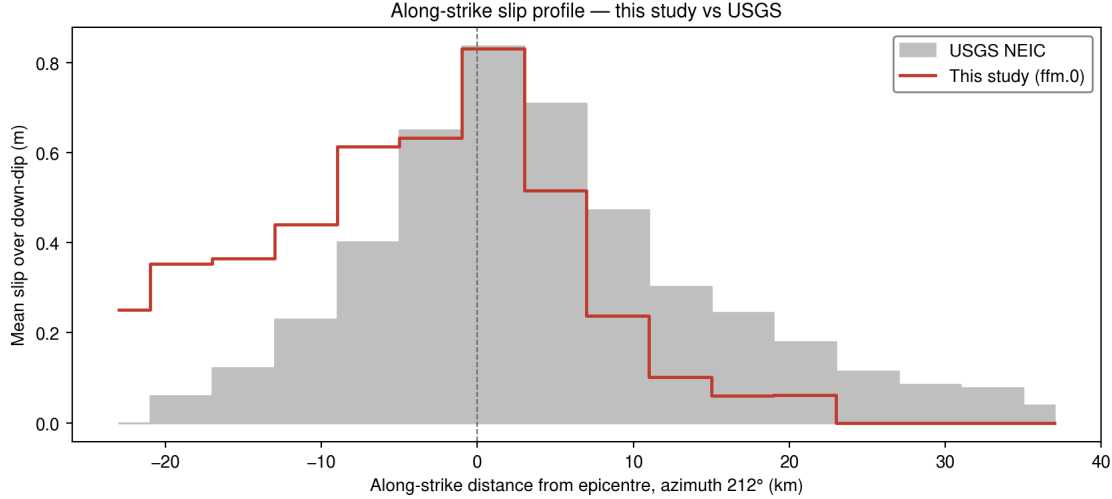


Figure 10: Along-strike slip profile (positive toward azimuth 212°): identical central asperity; the models disagree only on which side carries the low-slip tail.

Caveat on the agreement. WISP is the USGS NEIC operational code, so the parameterization family is shared: this blind test validates our data selection, Mww seeding, QC and plane handling rather than an independent inversion method (station sets and settings do differ: USGS used 42 body + 32 surface stations and 4 time windows). V_r differs by 20 % — teleseismic data constrain it weakly, consistent with the Mexico lesson.

7 Hypocentre depth and directivity

Hypocentre depth: 10 km (USGS origin, adopted by both inversions as the nucleation point; our WISP nucleation subfault 10.0 km, USGS on-fault 10.8 km, W-phase centroid 11.5 km) — near the bottom of the main slip patch (~ 2 –16 km). Moment-weighted centroids and early-vs-late rupture centroid motion, computed identically from both slip models:

metric	ours	USGS
moment NE vs SW of epicentre	57 % / 20 %	33 % / 56 %
centroid motion early→late, along-strike	$-1.9 \rightarrow -9.4$ km (NE)	$+0.6 \rightarrow +9.5$ km (SW)
centroid motion early→late, depth	8.3 → 17 km	7.8 → 12 km
moment updip vs downdip of hypocentre	43 % / 45 %	55 % / 32 %

Agreement (robust): the dominant moment-rate pulse (peak 3.4–3.7 s) breaks a compact asperity at and slightly updip of the hypocentre (early moment centroid ~ 8 km depth in both models) — a modest vertical (updip) directivity component, the classic nucleate-at-the-bottom, rupture-up pattern; thereafter rupture spreads bilaterally at low slip, with >75 % of the moment within ± 10 km of the epicentre.

Disagreement (therefore unresolved): the late low-slip tail goes NE in our model but SW in USGS’s — opposite senses of late centroid motion. When two inversions of the same data family put the directivity tail on opposite sides, horizontal directivity is not resolved. This is physically expected:

a ~ 14 s, ~ 20 – 30 km rupture produces azimuthal duration differences of only $\sim \pm 3$ – 4 s, marginal at >10 s teleseismic periods (unlike the Mexico M7.3 or the Venezuela doublet, where longer ruptures made directivity robust). Near-field strong motion (JMA/NIED) or InSAR would be needed to settle it.